MARS ROVER SAMPLE COLLECTION

PDDL Planning — Analysis and Goal Verification Report

Scenarios: Original five-rock mission and extended seven-rock mission with a dust-storm closure

Planning model	STRIPS + typing
Rover	Curiosity
Storage capacity	Two bays (Bay1, Bay2)
Equipment	One drill (Drill1)
Supplied plans analysed	Original: 25 actions | Extended: 40 actions
Verification	Both supplied plans simulated against the uploaded PDDL facts and action rules

Evidence note. This report analyses the PDDL files and plan text supplied with the request. The supplied material includes plans labelled optimal, but does not include a satisficing plan, planner configuration logs, expanded-state counts, or timing measurements. Those missing results are explicitly marked as unavailable; no planner statistics are fabricated.

1. Executive Summary

The domain models a rover moving through a directed connectivity graph, collecting soft and hard rocks into two capacity-limited bays, acquiring a drill, and depositing every rock at Base. The uploaded domain uses only **:strips** and **:typing**. The one-way slope is represented by the directed fact `(connected sitec sited)` without the reverse fact `(connected sited sitec)`. The movement action therefore respects route direction.

The supplied original plan contains **25 actions**; the supplied extended plan contains **40 actions**. Both were simulated successfully: all required rocks finish at Base, both bays are empty, and the rover finishes at Base. The extended plan is 15 actions longer (60% more actions), reflecting the two additional samples and the loss of the direct Base–SiteB connection.

> The plan files support verification of the two supplied plans, but they do not support a measured comparison between satisficing and optimal search. A valid comparison requires the missing satisficing plan and planner logs.

2. Domain and Problem Modelling Review

2.1 Types and objects

The domain declares three types: **location**, **rock**, and **bay**. The problem supplies seven locations (Base and SiteA–SiteF), two bays, and five rocks in the original scenario. The extended problem adds Rock6 and Rock7.

2.2 Predicates and their roles

Predicate	Meaning
rover-at(l)	Current rover location
connected(a,b)	Permitted directed movement edge
rock-at(r,l)	Rock r is physically at location l
drill-at(l), has-drill	Drill location and whether the rover has acquired it
soft(r), hard(r)	Rock collection requirement
bay-empty(b), holding(b,r)	Bay capacity and its stored rock

2.3 Action semantics

Move: requires the rover at the source and a directed connected edge; it changes the rover's location. **Pick-drill:** requires the rover and drill at the same location; it establishes `has-drill`, which persists. **Collect-soft:** requires a soft rock at the rover's location and an empty bay. **Collect-hard:** has the same conditions plus `has-drill`. **Deposit:** requires the rover at the deposit location and the selected bay holding that rock; it places the rock there and frees the bay.

2.4 Constraint coverage

The two bays enforce a maximum of two carried rocks because collection consumes `bay-empty` and deposit restores it. Rover/rock co-location is required by both collection actions. Hard-rock collection requires the drill. Deposit requires rover/bay co-location at the current location. Directed connectivity enforces legal movement. The goal explicitly requires every rock at Base, both bays empty, and the rover at Base.

The PDDL uses delete effects such as `(not (rover-at ?from))`; these are standard STRIPS effects, not negative preconditions. No numeric fluents or negative preconditions are used. The drill is picked up at SiteE and remains available because the `has-drill` fact is not deleted by any action.

3. Original Scenario: Plan Analysis

3.1 Plan summary

Action category	Count
Move	14
Pick Drill	1
Collect Soft	3
Collect Hard	2
Deposit	5

Total plan length: 25 actions (14 moves, 1 drill pickup, 3 soft collections, 2 hard collections, and 5 deposits).

3.2 Trip-by-trip interpretation

Trip 1 — hard rocks: The rover travels Base → SiteA → SiteC → SiteE, picks up the drill, then reaches SiteF and collects Rock5. It returns through SiteE and SiteC to SiteA, collects Rock3 at SiteC on the return route in the supplied plan sequence, and carries both hard rocks to Base for deposit. This trip uses both bays and avoids an unnecessary return to SiteF.

Trip 2 — SiteB and SiteD: From Base the rover visits SiteB, collects Rock2, moves to SiteD and collects Rock4 in the second bay, then returns via SiteB to Base and deposits both.

Trip 3 — SiteA: The rover visits SiteA, collects Rock1, returns to Base, and deposits it. The rover ends at Base with both bays empty.

3.3 Drill and bay strategy

The drill is acquired at SiteE before either hard rock is collected. In the first trip, Rock5 and Rock3 occupy Bay1 and Bay2 simultaneously. Both are deposited at Base before the next collection trip, freeing both bays. The second trip similarly carries Rock2 and Rock4 together. The final single-rock trip uses one bay. This batching reduces repeated travel to Base while respecting the two-rock capacity.

4. Extended Scenario: Plan Analysis

The extended problem adds Rock6 (soft) at SiteE and Rock7 (hard) at SiteF. The dust storm removes both directed Base–SiteB edges from the connectivity facts. The remaining route network still connects the rover to SiteB through SiteA, SiteC, SiteD, and the SiteB–SiteD link.

4.1 Plan summary

Action category	Count
Move	25
Pick Drill	1
Collect Soft	4
Collect Hard	3
Deposit	7

Total plan length: 40 actions (25 moves, 1 drill pickup, 4 soft collections, 3 hard collections, and 7 deposits).

4.2 How the rover reaches SiteB and SiteD

Because Base ↔ SiteB is closed, the rover cannot use the original direct connection. In the supplied plan it travels Base → SiteA → SiteC → SiteD, then reaches SiteB using the still-open SiteD → SiteB edge. It collects Rock4 at SiteD and Rock2 at SiteB, then returns SiteB → SiteD → SiteE → SiteC → SiteA → Base. Thus SiteB remains reachable despite the storm.

4.3 Use of the one-way slope

Yes. The extended plan uses the one-way edge **SiteC → SiteD** at action 16. This is legal because the PDDL contains that directed edge. The plan does not attempt SiteD → SiteC; it returns from SiteD through SiteB or SiteE instead. The slope is therefore used in its permitted direction.

4.4 Extended collection strategy

The first trip collects both hard rocks at SiteF (Rock5 and Rock7) after picking up the drill at SiteE, then returns to Base to unload. The next trip collects Rock4 at SiteD and Rock2 at SiteB, returning to Base through the alternate corridor. The next trip collects Rock3 at SiteC and Rock6 at SiteE, then unloads both at Base. A final short trip collects Rock1 at SiteA. This uses both bays on the first three trips and one bay on the final trip.

5. Plan Length Comparison and Search-Strategy Discussion

5.1 Results supported by the supplied plan files

Scenario	Provided plan label	Action count	Moves	Pick drill	Collections	Deposits
Original	Optimal path	25	14	1	5	5
Extended	Optimal path	40	25	1	7	7

The extended plan has 15 more actions than the original plan, a 60% increase. This is a cross-scenario comparison, not a comparison of search algorithms: the extended task has two additional rocks and fewer available edges. The extra travel is also visible in the increase from 14 to 25 move actions.

5.2 Satisficing versus optimal search: evidence limitation

A satisficing search (for example, greedy best-first search) aims to find a valid plan quickly but does not guarantee the minimum number of actions. An optimal search (for example, A* with an admissible heuristic or breadth-first search for unit-cost actions) guarantees a minimum-cost plan under the relevant assumptions. Therefore, a

satisficing plan may be longer, but it may be found with fewer expansions or less time. Actual results depend on the planner, heuristic, tie-breaking, and action-cost model.

Not present in the supplied evidence: no satisficing plan for either scenario, no planner name/configuration, no number of expanded states, and no planning-time measurements. Consequently, the satisficing-versus-optimal plan-length difference, expansions, and runtime cannot be reported or responsibly inferred. The two supplied plan files are treated as the optimal-labelled plans only; the report does not claim to have run an external planner.

For a complete experimental comparison, run both configurations on each problem with the same planner and action-cost assumptions, and record: plan length, expanded states, and wall-clock planning time. The resulting four planner runs (two strategies × two scenarios) should be included in the submission output.

6. Goal Verification by Plan Simulation

Each supplied action was replayed in order against the uploaded initial state, directed connectivity facts, and action preconditions/effects. Every action passed its applicable checks: movement used an existing directed edge; collection required rover/rock co-location and an empty bay; hard-rock collection occurred only after drill acquisition; and deposits were made at the rover's current location from the bay holding the relevant rock.

6.1 Final state — original scenario

Variable	Final value
Rover location	Base
Bay1	Empty
Bay2	Empty
Rock1–Rock5	All at Base
Drill	Carried by rover (has-drill is true)

Result: all original goal predicates are true. The original plan is valid with respect to the supplied domain and problem.

6.2 Final state — extended scenario

Variable	Final value
Rover location	Base
Bay1	Empty
Bay2	Empty
Rock1–Rock7	All at Base
Drill	Carried by rover (has-drill is true)

Result: all extended goal predicates are true. The plan obeys the storm-closed route and the one-way slope direction.

6.3 Final-state predicate checklist

Original: `rock-at(rock1..rock5, base)`, `bay-empty(bay1)`, `bay-empty(bay2)`, `rover-at(base)` — all true.

Extended: `rock-at(rock1..rock7, base)`, `bay-empty(bay1)`, `bay-empty(bay2)`, `rover-at(base)` — all true.

7. Conclusions

The uploaded PDDL model captures the key mission constraints: directed travel, two-bay capacity, soft/hard collection rules, drill acquisition, and delivery to Base. Both provided plans are valid when simulated against the supplied model. The original plan has 25 actions and the extended plan has 40 actions. In the extended scenario, the rover reaches SiteB by travelling through SiteA, SiteC, and SiteD; the one-way SiteC → SiteD slope is used legally.

The main outstanding deliverable is the empirical satisficing-versus-optimal comparison. To complete that part of the assignment, attach the satisficing plans and planner logs containing expanded-state counts and planning times. Until those are available, the report's search-strategy discussion is conceptual rather than a measured comparison.

Appendix A — Supplied plan action sequences

The complete action sequences are preserved in the supplied text files. The tables below show the ordered action list as used in verification.

A1. Original scenario — 25 actions

Step	Action
1	`(move base sitea)`
2	`(move sitea sitec)`
3	`(move sitec sitee)`
4	`(pick-drill sitee)`
5	`(move sitee sitef)`
6	`(collect-hard rock5 sitef bay1)`
7	`(move sitef sitee)`
8	`(move sitee sitec)`
9	`(collect-hard rock3 sitec bay2)`
10	`(move sitec sitea)`
11	`(move sitea base)`
12	`(deposit rock5 base bay1)`
13	`(deposit rock3 base bay2)`
14	`(move base siteb)`
15	`(collect-soft rock2 siteb bay1)`
16	`(move siteb sited)`
17	`(collect-soft rock4 sited bay2)`
18	`(move sited siteb)`
19	`(move siteb base)`
20	`(deposit rock2 base bay1)`
21	`(deposit rock4 base bay2)`
22	`(move base sitea)`
23	`(collect-soft rock1 sitea bay1)`
24	`(move sitea base)`
25	`(deposit rock1 base bay1)`

A2. Extended scenario — 40 actions

Step	Action
1	(move base sitea)
2	(move sitea sitec)
3	(move sitec sitee)
4	(pick-drill sitee)
5	(move sitee sitef)
6	(collect-hard rock5 sitef bay1)
7	(collect-hard rock7 sitef bay2)
8	(move sitef sitee)
9	(move sitee sitec)
10	(move sitec sitea)
11	(move sitea base)
12	(deposit rock5 base bay1)
13	(deposit rock7 base bay2)
14	(move base sitea)
15	(move sitea sitec)
16	(move sitec sited)
17	(collect-soft rock4 sited bay1)
18	(move sited siteb)
19	(collect-soft rock2 siteb bay2)
20	(move siteb sited)
21	(move sited sitee)
22	(move sitee sitec)
23	(move sitec sitea)
24	(move sitea base)
25	(deposit rock4 base bay1)
26	(deposit rock2 base bay2)
27	(move base sitea)
28	(move sitea sitec)
29	(collect-hard rock3 sitec bay1)
30	(move sitec sitee)
31	(collect-soft rock6 sitee bay2)
32	(move sitee sitec)
33	(move sitec sitea)
34	(move sitea base)
35	(deposit rock3 base bay1)
36	(deposit rock6 base bay2)
37	(move base sitea)
38	(collect-soft rock1 sitea bay1)
39	(move sitea base)
40	(deposit rock1 base bay1)